Independent exact reproduction of the sign-dependent teleportation signal in a chaotic sparse-SYK traversable-wormhole protocol

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Abstract

We independently reproduce, by exact statevector simulation, the central qualitative signature of Byun, Kim and Lee [1] (arXiv:2604.10090): a *sign-dependent asymmetry* in the reference–target mutual information of a traversable-wormhole (TW) teleportation protocol on a chaotic binary sparse Sachdev–Ye–Kitaev (SYK) model. At $N = 8$, $q = 4$, $\beta = 3$, $|\mu| = 12$, in the fixed-injection protocol, we obtain a peak $\Delta I_{PT} \equiv I_{PT}^{\mu<0} - I_{PT}^{\mu>0} = +0.0168$ bits at $t_1 = 2.0$ ($I_{PT}^{\mu<0} = 0.0189$, $I_{PT}^{\mu>0} = 0.0021$), positive as the paper reports. The requisite spectral chaos reproduces the expected random-matrix behaviour, including the N mod 8 Bott periodicity of the level-spacing ratio; we are explicit that at $N = 8$ the chaos backing is *relative* (sparse matches dense SYK_4) rather than an absolute RMT-vs-Poisson certification. Crucially, an independent analytical audit established that reproducing an asymmetry of the right *magnitude* does not by itself confirm the paper's *physical* sign; we therefore verified the sign convention directly against the paper's Eq. (3), which fixes $\mu < 0$ as the ANEC-violating (traversable) sign in exactly our operator convention. The reproduction is faithful in sign, magnitude, and spectral chaos. We do not reproduce the paper's NISQ-hardware run; this is the exact-simulation baseline against which such a run is judged.

1 What we reproduced

The TW teleportation protocol [1, 2, 3] evolves a two-sided SYK state and reads out a mutual-information signal:

$$|\mathrm{out}\rangle = \mathrm{SWAP}_{RT}\, U(t_1)\, e^{i\mu V}\, U(t_0)\, \mathrm{SWAP}_{QL}\, U(-t_0)\, |\mathrm{in}\rangle, \qquad U(t) = e^{-i(H_L+H_R)t}, \tag{1}$$

with the instantaneous fermionic bilinear coupling

$$V = \frac{1}{qN}\, H_{\mathrm{int}}, \qquad H_{\mathrm{int}} = i\sum_{j=1}^{N} \psi_L^j \psi_R^j, \tag{2}$$

and the teleportation signal

$$I_{PT}(t_0, t_1) = S_P + S_T - S_{PT}, \tag{3}$$

the base-2 von Neumann entropies of the reduced density matrices of the reference P, target T, and their union. We use the paper's parameters $N = 8$, $q = 4$ (binary sparse, K retained terms), $\beta = 3$, $|\mu| = 12$, and the fixed-injection protocol (t_0 fixed, t_1 swept). Our implementation is exact statevector arithmetic (Jordan–Wigner Majoranas with $\{\gamma_i, \gamma_j\} = 2\delta_{ij}$; left/right factors made to anticommute via a parity dressing), reproduced by `labs/track-d-magic/phase6_tw_protocol.py` and `phase6_chaos_diagnostics.py` in the repository.

2 Results

Sign-dependent asymmetry (the signal). In the fixed-injection protocol we find the mutual-information transfer peaks near $t_1 = 2.0$ with

$$I_{PT}^{\mu<0} = 0.0189, \quad I_{PT}^{\mu>0} = 0.0021, \quad \Delta I_{PT} = +0.0168 \text{ bits}, \tag{4}$$

i.e. the signal is enhanced for $\mu < 0$, matching the paper's reported positive $\Delta I_{PT} \equiv I_{PT}^{\mu<0} - I_{PT}^{\mu>0}$.

Spectral chaos (the prerequisite). A wormhole reading is meaningful only if the sparse model retains the spectral chaos of the dense SYK; this is the exact point the 2019–2023 debate over learned-Hamiltonian "wormhole" claims turned on. The level-spacing ratio $\langle r \rangle$ reproduces the SYK Bott periodicity by N mod 8: $N = 8$ GOE-like ($\langle r \rangle \approx 0.53$), $N = 10$ GUE-like (≈ 0.60), $N = 12$ GSE-like (≈ 0.67); the free SYK_2 control trends Poisson (≈ 0.39–0.47). The Gaussian-filtered spectral form factor of the sparse $K \approx 10$ ensemble matches the dense $K = 70$ reference. *Honest bound:* at the accessible $N = 8$ the even-parity spectrum is too small for an absolute RMT-vs-Poisson certificate; a synthetic Poisson sequence also shows a ramp. The $N = 8$ chaos claim is therefore *relative* (sparse $\approx$ dense), with absolute chaos certification only at $N \geq 10$ — a limitation we share with, and do not paper over relative to, the source work.

3 Sign-convention faithfulness (why an independent audit mattered)

A magnitude-matching asymmetry is not sufficient: reproducing $|\Delta I_{PT}|$ while inverting the coupling sign would reproduce the number and invert the physics (the "negative-energy shockwave" would be a positive-energy one). An independent analytical review flagged that $\mu < 0$ being the traversable sign is *not* forced by $e^{i\mu V}$ and our V alone; it requires the paper's explicit convention or the standard Maldacena–Qi / Gao–Jafferis–Wall double-trace sign for negative null energy [2, 3]. We resolved this directly: the paper states, in the operator convention it adopts (Eq. (3) and surrounding text), that "$\mu < 0$ corresponds to ANEC violation and hence to traversability." Our $V = H_{\text{int}}/qN$ with $H_{\text{int}} = i \sum_j \psi_L^j \psi_R^j$ and coupling $e^{+i\mu V}$ is term-for-term the paper's, and our ΔI_{PT} is defined with the same sign. The reproduction is therefore faithful in *sign*, not merely in magnitude.

4 Scope and limitations

- **Exact simulation, not hardware.** We reproduce the exact-statevector baseline. The paper's distinct contribution is a NISQ-hardware demonstration of the same qualitative signature under noise; that we do not reproduce.
- **Relative chaos at $N = 8$.** As above: sparse $\approx$ dense at $N = 8$; absolute RMT certification needs $N \geq 10$.
- **Qualitative, not quantitative gravity.** As the authors themselves stress, a sign-dependent mutual-information asymmetry in a finite-N chaotic model is a holographically *motivated* signature, not evidence of literal bulk gravitational dynamics.

5 Methods — three-model adversarial verification

This reproduction was produced by three AI research agents on distinct model families, each in the role its capabilities fit: execution and numerics (codex-science/Tobin, OpenAI), analytical/sign-convention audit (agy-science, Google), and synthesis and source verification (quant-phy/Ariadne, Anthropic), under human direction. The sign-faithfulness check — the one point at which a magnitude-only reproduction could have silently passed on a flipped convention — came from the non-executing analytical reviewer, not the execution pass. Every number is a commit hash or a recorded run in the public repository; the source paper's parameters and sign convention are quoted, not inferred.

References

[1] S. Byun, J. Kim, J. Lee, *Quantum simulation of traversable-wormhole-inspired quantum teleportation in a chaotic binary sparse SYK model*, arXiv:2604.10090.

[2] J. Maldacena, X.-L. Qi, *Eternal traversable wormhole*, arXiv:1804.00491.

[3] P. Gao, D. L. Jafferis, A. C. Wall, *Traversable Wormholes via a Double Trace Deformation*, arXiv:1608.05687.

Acknowledgements. Produced by a swarm of AI research agents — a mix of OpenAI, Google, and Anthropic models — under human direction, each independently checking the others before a claim was allowed to stand. Code and data are public; every computational claim carries a commit hash or a recorded run.